Yuanbo Pang

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Dear Astera Institute Hiring Team,

I am excited to apply for the Software Engineering Intern, Distributed Simulation Systems role. Astera's focus on building high-velocity scientific infrastructure resonates with the work I have been doing at the intersection of agent evaluation, robotics perception, and reliable engineering systems.

At UC Berkeley BAIR, I help build infrastructure for Agent's Last Exam, a NeurIPS 2026 benchmark submission for evaluating frontier AI agents on long-horizon professional workflows. My work involves converting expert-submitted tasks into reproducible VM-based GUI/CLI benchmark runs with artifact capture and deliverable-based scoring. That experience maps closely to Astera's need for simulation tooling that is measurable, debuggable, and useful to researchers.

I also bring robotics and perception context from my work with Prof. Avideh Zakhor on UAV small-object detection and tracking. I have studied leading UAV detection methods and evaluated motion-based cues on 1,600 manually annotated DJI frames, comparing raw frame differencing against optical-flow and background-compensation approaches. This has made me especially interested in simulation systems where physics, vision, and evaluation infrastructure meet.

Beyond research, I have shipped real software systems end to end. At Geopogo, I was the sole engineer on a production AI rendering platform built with React, TypeScript, FastAPI, Firebase, Vercel, Gemini API, and RunwayML. At Starbot, I solo-built an agent harness for restaurant service robots that coordinated ordering, delivery, and customer-service workflows for a CES 2026 demo covered by NHK News.

I would be excited to contribute to Astera's distributed simulation infrastructure, internal tooling, and visualization workflows while learning from engineers and researchers building ambitious scientific systems. Thank you for your consideration.

Sincerely,
Yuanbo Pang